A Survey on IoT-Based Underground Cable Fault Detection Systems

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***Abstract*—Underground cable systems are widely used in modern power distribution networks due to their improved reliability and protection from environmental disturbances. However, detecting and locating faults in underground cables remains a challenging task because the cables are buried and cannot be visually inspected. This survey analyzes five research papers published between 2021 and 2025 that focus on underground cable fault detection using signal processing, deep learning, and IoT-based monitoring systems. The methodologies, datasets, fault types, challenges, and research gaps are examined to understand current trends and future research directions.**

I. Introduction

Underground power cables play a crucial role in urban power distribution systems. Compared to overhead lines, underground cables provide better safety, reduced visual pollution, and improved reliability. However, faults in underground cables are difficult to detect due to their hidden installation. Traditional techniques such as the Murray loop test and thumping method require manual intervention and are time-consuming.

With the advancement of Internet of Things (IoT), signal processing, and machine learning technologies, researchers have proposed intelligent monitoring systems that enable real-time fault detection and remote monitoring. This survey reviews recent contributions in this domain.

II. Literature Review

The 2021 research proposed a similarity-based framework for detecting incipient faults using voltage waveform analysis. The system utilized cross-correlation and curve fitting techniques to distinguish fault signals from normal disturbances.

The 2022 study introduced a deep learning-based approach using 1D Convolutional Neural Networks (1D-CNN) and Support Vector Machine (SVM) to classify multiple types of underground cable faults with high accuracy.

The 2020 research presented a basic IoT-based underground cable fault detector using Arduino and resistance measurement techniques to estimate fault location.

The 2024 paper implemented an IoT-based monitoring system using GSM modules and current sensors for real-time alert generation.

The 2025 study proposed a smart monitoring system integrating NodeMCU and cloud platforms for predictive and remote cable monitoring.

III. Methodology

The methodologies adopted in the reviewed papers can be categorized into three types:

A. Signal Processing-Based Approach

The 2021 study used voltage waveform similarity analysis and cross-correlation to detect incipient faults. This method analyzes changes in signal patterns to identify abnormalities.

B. Machine Learning-Based Approach

The 2022 research employed deep learning techniques such as 1D-CNN combined with SVM for fault classification. Feature extraction methods were applied to improve classification performance.

C. IoT-Based Monitoring Approach

The 2020, 2024, and 2025 studies implemented real-time monitoring systems using sensors (current, voltage), microcontrollers (Arduino, NodeMCU), and communication modules (Wi-Fi, GSM). When abnormal values exceed threshold limits, alerts are generated and transmitted to cloud platforms.

IV. Dataset Analysis

The datasets used in the reviewed research vary based on methodology. The 2021 and 2022 studies primarily relied on simulated datasets generated using software tools such as ATP–EMTP and MATLAB. These datasets include voltage and current transient signals.

The IoT-based studies used real-time sensor-generated datasets collected from hardware prototypes. These datasets consist of current readings, voltage measurements, resistance values, and time-stamped monitoring data.

A major observation is that none of the studies used a standardized large-scale public dataset for underground cable faults. Most data were either simulated or collected from small-scale experimental setups.

V. Types of Faults

The reviewed papers addressed several fault types:

- Short Circuit Fault – Occurs when two conductors come into contact.
- Earth Fault – Occurs when a conductor touches the ground.
- Open Circuit Fault – Occurs due to conductor breakage.

- High Resistance Fault – Produces low current but can be dangerous.
- Incipient Fault – Weak temporary fault caused by insulation degradation.

Detecting incipient and high-resistance faults requires advanced signal analysis techniques.

VI. Comparative Analysis

The comparative analysis shows a transition from traditional signal-based techniques to intelligent machine learning and IoT-based monitoring systems. Deep learning models provide high classification accuracy but depend heavily on simulated data. IoT-based systems offer practical real-time monitoring but require reliable communication networks.

VII. Challenges and Research Gaps

Despite advancements, several challenges remain:

- Lack of standardized real-world datasets
- Limited large-scale field deployment
- Internet dependency in IoT systems
- Cybersecurity concerns

Future research should focus on integrating machine learning with real-time IoT data for predictive maintenance.

VIII. Conclusion

This survey reviewed five research papers from 2021 to 2025 on underground cable fault detection systems. The study demonstrates the evolution from signal processing techniques to deep learning and IoT-based smart monitoring systems. Although significant progress has been made, standardized datasets and large-scale deployment remain important research challenges.

IX. Comparative Discussion

The comparative evaluation of the reviewed studies reveals a gradual evolution in underground cable fault detection methodologies. The 2021 research primarily relied on signal processing techniques, focusing on waveform similarity analysis to detect incipient faults. Although effective for identifying subtle disturbances, the approach depends heavily on signal quality and may not perform well under severe noise conditions.

The 2022 deep learning-based study significantly improved fault classification accuracy by leveraging feature extraction and 1D convolutional neural networks. While this approach achieved high precision in simulated environments, its dependency on synthetic datasets limits real-world validation.

The IoT-based studies from 2020, 2024, and 2025 demonstrate practical implementation strategies by integrating sensors, microcontrollers, and communication modules. These systems enable real-time monitoring and remote alert mechanisms. However, they often lack advanced predictive algorithms and large-scale field testing.

Overall, signal processing techniques provide analytical insight, machine learning enhances classification accuracy, and IoT systems ensure real-time deployment. A hybrid integration of these three methodologies represents the most promising future direction.

X. Future Scope

Future research should emphasize the development of standardized underground cable fault datasets collected from real-world distribution systems. Integrating deep learning models with real-time IoT monitoring platforms can enable predictive maintenance rather than reactive fault detection.

Furthermore, incorporating cybersecurity mechanisms into IoT communication frameworks is essential to prevent data breaches and system vulnerabilities. The adoption of edge computing for on-site data processing can also reduce latency and improve fault response time.

XI. Enhanced Conclusion

This survey comprehensively reviewed recent advancements in underground cable fault detection systems between 2021 and 2025. The study highlights a clear transition from conventional analytical approaches to intelligent machine learning models and IoT-enabled smart monitoring systems. While deep learning improves classification accuracy and IoT enhances real-time monitoring capabilities, the absence of standardized datasets and large-scale validation remains a major research challenge. Addressing these issues will significantly improve the reliability and scalability of future underground cable monitoring systems.

section*References

1) H. Samet, S. Khaleghian, M. Tajdinian, T. Ghanbari, and V. Terzija, "A similarity-based framework for incipient fault detection in underground power cables," *International Journal of Electrical Power & Energy Systems*, vol. 133, 2021.
2) A. Said, S. Hashima, M. M. Fouda, and M. H. Saad, "Deep Learning-Based Fault Classification and Location for Underground Power Cable," *IEEE Access*, vol. 10, pp. 70134–70142, 2022.
3) K. R. Kavitha, S. Vijayalakshmi, V. B. Kripal, D. Sathishkumar, K. Kishorekumar, and S. Dinesh, "IoT Based Underground Fault Detector," *Journal of Complementary Medicine Research*, vol. 11, no. 1, 2020.
4) V. Hari Krishna, G. Neelima, P. Poojitha, M. Madhu, D. V. N. Surya, and Ch. Satwika, "Underground Cable Fault Detection Using IoT," *International Journal of Novel Research and Development*, vol. 9, no. 5, 2024.
5) S. Bharathi, K. Harini Sri, K. Deepika Sri, and S. Dhaarini, "Smart Underground Fault Monitoring and Detection Using Internet of Things," in *Proc. ISPES*, 2025.